Question				Marking details	Marks Available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
1	*(a)*			Ticks in 1st, 2nd and 3rd boxes 3 × (1) 4 boxes ticked maximum = 2 marks 5 boxes ticked maximum = 1 mark 6 boxes ticked = 0 marks	3			3		
	(b)			Beta emitters are most suitable as beta would be partly absorbed [Accept: beta will pass through thin Al / blocked by a few mm [or thick] Al] / alpha totally absorbed and gamma not absorbed (1) Increasing thickness decreases beta count rate [accept: changing thickness would change count rate] (1) Sr-90 has a long enough half-life so won't need frequent replacing / P-32 would need frequent replacing. (1) For 3 marks "agree" required.			3	3		
	(c)	(i)		It takes 29 years / this is the time to halve (1) number of nuclei / atoms / mass / amount / activity / count rate [of strontium-90] (1)	2			2		
		(ii)		[1 →] $\frac{1}{2}$ → $\frac{1}{4}$ → $\frac{1}{8}$ or [100] →50→25→12.5% (1) multiple halving ending with $\frac{1}{8}$ (12.5%) so $\frac{1}{8}$ is 3 half-lives **ecf** on incorrect halving or incorrect counting of half-lives (1) 3 **ecf** × 29 = 87 years (1) NB 87 years → 3 marks; 58 or 116 years → 2 marks		3		3	3	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		150 ÷ time (1) = 0.5 [cps] (1) ans		2		2	2	2
		(ii)		Measure for a longer period of time (1) take repeat readings (1)			2	2		2
				Question 1 total	**5**	**5**	**5**	**15**	**5**	**4**